



GRID DOWN

GRID DOWN DATA BLUEPRINT

Field Manual for Offline Knowledge Operations

CLASSIFICATION: AIR-GAPPED // OFFLINE // FOR FIELD USE ONLY

EDITION 1.0 – 2026

DOC. NO. **GDB-001**

FOR HOLDER USE

RETAIN WITH DATA DRIVE

GRID DOWN DATA // AIR-GAPPED SURVIVAL VAULT

DOCUMENT CONTROL

0.1 PURPOSE

This manual governs the storage, activation, and field use of the Grid Down Data Drive. It assumes total loss of internet, cellular, and grid power. Every procedure herein is executable with the drive, one charged device, and the contents of this document.

0.2 HOW TO USE THIS MANUAL

— DIGITAL TIER HOLDER PROTOCOL — START HERE

If you purchased this Blueprint as a standalone digital asset and do not possess a preloaded physical drive, **DO NOT proceed to Section 1**. Stop reading here and immediately skip to **SECTION 8: DIY VAULT BUILD COMPENDIUM** at the back of this document. You must execute the hardware sourcing and data-staging procedures to construct your offline vault before these operational protocols can be utilized.

Read §1 and §2 before an emergency. Execute §2 during one. Sections §3–§6 are reference protocols — consult them in priority order: **water, navigation, medical, communications**. §7 contains recurring checklists; run them on schedule, not from memory.

0.3 PRIORITY LEGEND

MARKER	MEANING
IMMEDIATE	Execute within first 30 minutes of activation
72-HR	Complete within the first three days
SUSTAINED	Recurring action for the duration of the event

0.4 REVISION BLOCK

Verify you hold the current edition during each quarterly maintenance cycle (§7.2). Superseded editions: destroy or mark VOID across the cover.

REV	DATE	CHANGE SUMMARY
1.0	2026	Initial issue

▲ SCOPE LIMITATION — 0.5

This manual is a decision aid, not a substitute for training or professional medical care. Where licensed care is reachable, seek it. Procedures reflect accepted emergency-management practice at time of issue.

THE ASSET: YOUR DATA DRIVE

1.1 CONTENT MANIFEST

The drive carries a curated offline library in four top-level directories. Directory names on the drive match this table exactly. Verify quarterly (§7.2).

DIRECTORY	CONTENTS	PRIMARY USE
00_README_and_Blueprint	This manual, quick-reference guides, dashboard orientation assets	Start here; §0–§2
01_Kiwix_Offline_Wikis	The WikiMed medical encyclopedia and general reference library, stored as compressed offline archive files (ZIM format) readable without any internet	Symptom and technical lookup (§5)
02_Custom_Survival_Checklists	Printable action lists; water/medical/food checklists; maintenance logs	§3, §5, §7 execution
03_Offline_Maps	OsmAnd map regions, elevation and terrain map sets, and map tile databases (MBTiles format)	Navigation and route planning (§4)

1.2 HARDWARE SPECIFICATION AND HANDLING LIMITS

The drive is a ruggedized USB unit, fully waterproof (IP68 rated — it survives complete submersion), operating from –20 °C to 60 °C (–4 °F to 140 °F).

- Do **not** leave it connected to an unattended powered device.
- Do **not** expose the connector to salt water.
- Do **not** store above 60 °C — a vehicle dashboard in the sun exceeds this.

■ DOCTRINE

The shell survives abuse. The data survives discipline.

1.3 FARADAY POUCH PROTOCOL

The silver pouch is protective electromagnetic shielding — it blocks the kind of energy surge that can destroy electronics — not packaging.

- **Sealed state (default):** Drive lives inside the pouch, pouch fully closed with a double fold. Storage location: cool, dry, known to all household members.
- **Deployed state:** Drive is out for use or scheduled maintenance. Minimize deployed time during solar-storm warnings or heightened threat conditions.

△ RE-SEAL PROCEDURE — EVERY TIME

Wipe the drive dry, confirm cap seated, insert, double-fold the closure, verify no gap at the fold. **A pouch that does not close completely provides no protection.**

1.4 REDUNDANCY DOCTRINE — THE 2×2 RULE

One copy is zero copies. Maintain **two drives** in **two locations** (residence + vehicle, or residence + secondary site).

- Both copies follow the same maintenance schedule (§7.2).
- If one copy is deployed for use, the second remains sealed.
- Never transport both copies in the same container.

■ STEADY STATE = READINESS

Everything in this section is executed **before** an event. During an event you verify; you do not prepare.

ACTIVATION PROCEDURE (FIRST 30 MINUTES)

2.1 DECISION TRIGGER

IMMEDIATE

Break the Faraday seal when **any two** of the following conditions persist beyond 2 hours:

#	INDICATOR	VERIFY BY
1	Grid power loss across your area, not just your structure	Line of sight to neighboring structures / streetlights
2	Cellular failure — voice and data — on all household carriers	Two devices, two carriers if available
3	Internet failure on every household path (fixed, mobile, tethered)	Router status + mobile hotspot test
4	Official emergency broadcast declaring regional disruption	Battery/hand-crank radio, NOAA or local emergency broadcast

▲ ACTIVATION RULE

One indicator alone is an outage — **hold the seal**. Two sustained indicators are an event — **activate**. Record time of activation on the §7 maintenance log.

2.2 DEVICE POWER AUDIT AND LOAD ORDER

IMMEDIATE

Before connecting anything, inventory every charged device and record battery percentages. Then assign roles:

- 01 Primary terminal** — the phone with the highest charge. All drive operations run here. Airplane mode ON, GPS ON, brightness ≤ 30%, auto-lock 30 seconds.
- 02 Reserve terminal** — second-best device. Power off completely — not sleep, off. Untouched until the primary fails or a scheduled power window opens.
- 03 Tier three** — laptops. Highest capability, worst consumption. Deploy only for tasks phones cannot do (bulk file copy, large map regions).

2.3 CONNECTING THE DRIVE

IMMEDIATE

- **Android:** Connect via USB-C (adapter stored in the pouch). Acknowledge the storage permission prompt. File manager → drive volume → 00_README_and_Blueprint → START_HERE.html.
- **iOS:** Connect via Lightning or USB-C adapter. Files app → Locations → drive volume → same path, same entry point.
- **Laptop:** The drive appears like any ordinary USB stick. Open START_HERE.html in any browser. No installation, no network, no special permissions.

▲ MOUNT FAILURE DRILL — ONE VARIABLE AT A TIME

(1) swap cable → (2) swap port → (3) swap device. If all three fail, the drive goes back in the pouch and the reserve terminal is tried at the next power window. **Do not troubleshoot past 10 minutes on the primary terminal.**

2.4 DASHBOARD ORIENTATION

72 - HR

START_HERE.html opens the offline dashboard. Confirm the four tiles against Fig. 2-A. The search bar queries local content only. Four confirmed tiles = system nominal; proceed to §3.

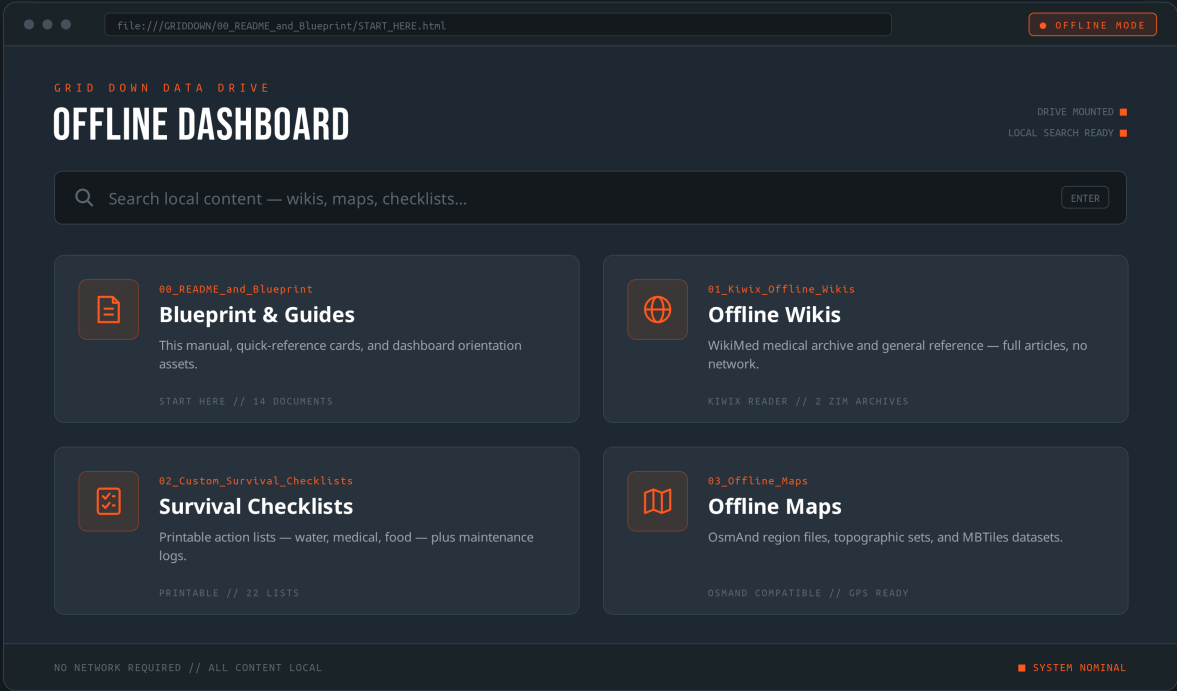


FIG. 2-A // OFFLINE DASHBOARD — START_HERE.HTML

FIG. 2-A // OFFLINE DASHBOARD — START_HERE.HTML // TILES MIRROR DRIVE DIRECTORIES 00-03

SECTION 3

WATER PROCUREMENT, PURIFICATION & SHORT-TERM FOOD RATIONING

3.1 SOURCE HIERARCHY AND DAILY REQUIREMENTS

72-HR

Plan on **4 liters (about 1 gallon) per person per day** — 2 drinking, 2 for food preparation and hygiene. Hot climate or physical work: raise drinking allocation to 3–4 liters. Procure in this order of preference:

PRIORITY	SOURCE	TREATMENT REQUIRED
1	Sealed commercial water; water heater tank (30–50 gal); the clean water storage tank mounted behind the toilet — never the bowl	None if sealed / known-clean plumbing
2	Rainwater collected mid-storm off clean surfaces	Filter + bleach
3	Flowing surface water — streams, rivers (collect upstream of settlements)	Full three-barrier
4	Standing water — ponds, lakes, pools	Full three-barrier; avoid if alternatives exist
5	Flood water, water near roads/agriculture/industry	Last resort — full three-barrier; chemical contamination may survive all treatment

■ ACTIVATION-DAY TASK

Drain the water heater: close the inlet valve, open a hot tap to break the vacuum, drain from the base spigot through cloth. 30–50 gallons of clean water most households forget they have.

3.2 THE THREE-BARRIER STANDARD

SUSTAINED

Full treatment for any untrusted source runs all three barriers in order. Skip a barrier only when the table in 3.1 permits.

- 01 **Barrier 1 — Filter.** Pre-filter through cloth to remove dirt and debris, then through your primary filter — a high-efficiency water filter rated to strain out bacteria and parasites. Filtering does not remove viruses or chemicals.
- 02 **Barrier 2 — Bleach.** Ordinary, unscented household bleach: **2 drops per liter** of clear water, **4 drops per liter** of cloudy water. Mix, wait **30 minutes**. A faint chlorine smell means the dose is correct. No smell = re-dose once and wait again. Purification tablets per the package label are an equivalent substitute.
- 03 **Barrier 3 — Boil.** Rolling boil for **1 minute** (3 minutes above 2,000 m / 6,500 ft). The final barrier when fuel permits — boiling kills everything biological.

▲ FUEL - SCARCE PROTOCOL

Barriers 1 + 2 only, reserving boiled water for the sick, infants, and wound care. Printable sequence:

02_Custom_Survival_Checklists → water checklist.

3.3 FIELD-EXPEDIENT COLLECTION

SUSTAINED

- **Rain:** Tarp or poncho angled into a container; discard the first 10 minutes of runoff.
- **Dew:** Absorb from clean grass or metal at dawn with cloth, wring into a container. Low yield — supplement only.
- **Tree-bag method:** Clear plastic bag sealed over a leafy branch, low corner weighted; collects moisture through daylight. Yield ~100–300 ml per bag per day. Non-toxic species only.
- **Snow/ice:** Melt before consuming — eating snow lowers body temperature. Treat as Priority 2.

■ NEVER COLLECT FROM

Vehicle radiators · waterbeds · chemically treated pools (unknown dosing) · any source with dead animals, an oily sheen, or a sewage odor upstream.

3.4 STORAGE ROTATION AND CONTAINERS

SUSTAINED

- Store treated water in food-grade containers, filled to the top (minimal air), labeled with treatment date and method.
- Rotate on a **6-month cycle** in steady state; during an event, oldest-treated is consumed first.
- Keep treated and untreated containers physically separated and marked — a single mix-up defeats the entire barrier system.
- Minimum standing reserve: **72 hours × household headcount × 4 L.**

3.5 EMERGENCY CALORIE MANAGEMENT AND FOOD ASSET PROTECTION

SUSTAINED

- **Ration baseline:** Healthy adults function on **1,200–1,500 kcal/day** for 72 hours without performance collapse. Children, pregnant women, and the ill are exempt — feed to need.
- **Consumption order:** (1) refrigerated — first 24 hours; (2) frozen — days 2–3 as thaw allows (a full, unopened freezer holds safe temperature ~48 hours); (3) shelf-stable pantry; (4) dedicated emergency stores last. Keep refrigerator and freezer doors closed; every open-and-shut cycle slashes your safe window by hours.
- **Asset protection:** Consolidate shelf-stable stores into sealed, rodent-proof containers in the coolest available space. Inventory on day 1; log consumption daily on the §7 checklist. Utilize the printable calorie tracking logs in [02_Custom_Survival_Checklists](#) to preserve device battery life. Do not advertise stores; food security is operational security.
- **Cooking discipline:** Batch-cook once daily to conserve fuel; cold-soak options (oats, instant rice) preserve fuel for water treatment.

▲ THE 4-HOUR RULE

Refrigerated perishables above 4 °C / 40 °F for more than 4 hours are discarded. Food poisoning in a grid-down environment is a casualty you cannot afford — **when in doubt, throw it out.**

■ FUEL DOCTRINE

Water outranks food for every fuel decision.

TACTICAL NAVIGATION & OFFLINE MAPS

4.1 OSMAND SETUP

IMMEDIATE

The drive carries everything required — installer included. No network is used at any step. Execute the three-step sequence in Fig. 4-A; verify your position pin sits on the correct street or terrain feature before trusting any route.

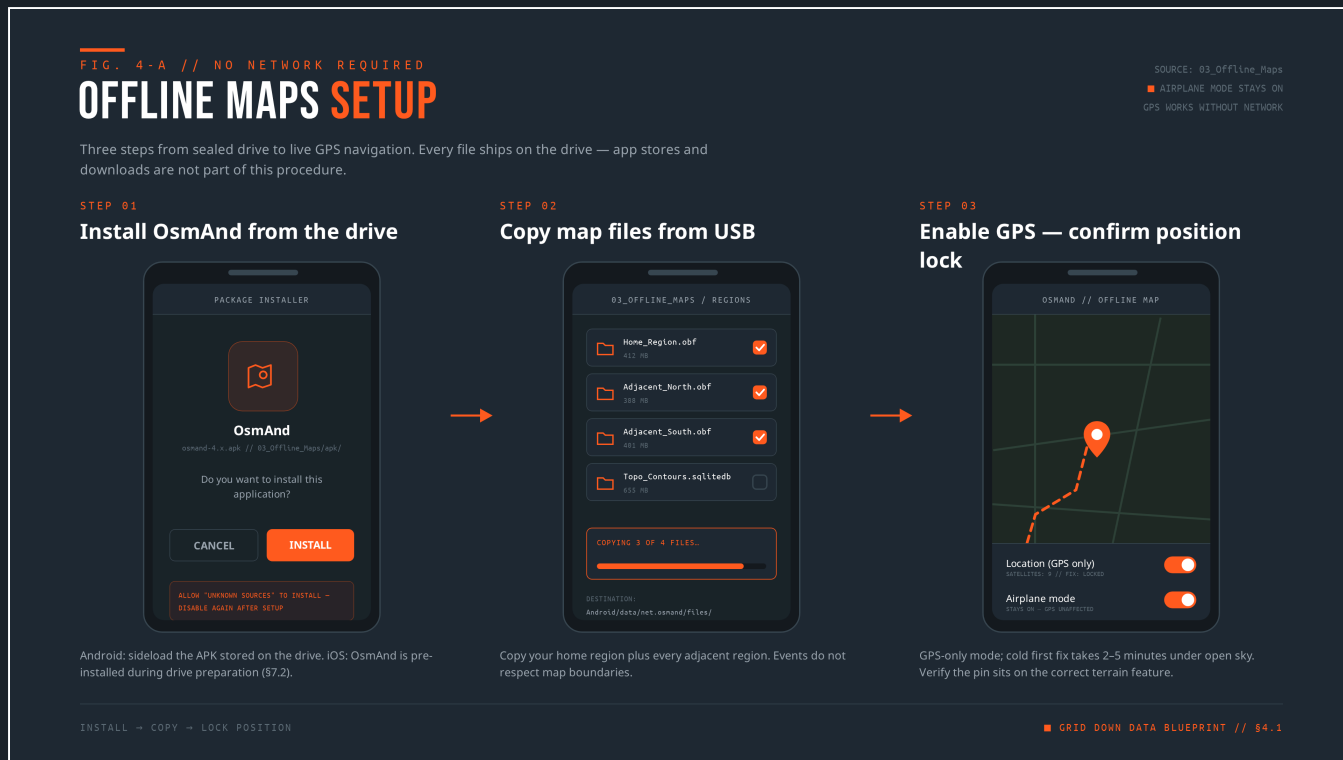


FIG. 4-A // OFFLINE MAPS SETUP — INSTALL FROM DRIVE → COPY REGIONS → CONFIRM GPS POSITION

⚠️ iOS PREREQUISITE — STEADY STATE

iOS users must download the free OsmAnd Maps app from the App Store during initial drive setup while internet is still available; app stores do not exist in a grid-down environment (§7.2 verifies quarterly). Android installs offline from `03_Offline_Maps/apk/` — allow "install from unknown sources," then switch that permission back off after setup.

4.2 GPS DISCIPLINE

SUSTAINED

GPS reception is free; the screen is not. Navigate from memorized route legs, waking the screen for a position check at **decision points only** — junctions, terrain changes, every 20–30 minutes on open ground. Leaving live navigation running drains a phone in hours; disciplined position checks stretch it to days. Log daily start/end positions on paper.

4.3 ROUTE PLANNING — PACE

72 - HR

Every movement gets four routes planned before departure:

ROUTE	STANDARD
PRIMARY	Fastest safe route; avoids chokepoints (bridges, tunnels, underpasses) where possible
ALTERNATE	Reaches the same objective sharing none of Primary's chokepoints
CONTINGENCY	Fallback if the objective itself is compromised — routes to secondary objective
EMERGENCY	Shortest path to nearest safe haven from the midpoint of any route above

■ PAPER DOCTRINE

Plan on the device; transcribe turn lists to paper. **The paper copy is the primary navigation document; the device is the verification tool.**

4.4 MARKING WATER, MEDICAL, AND RALLY POINTS

SUSTAINED

Use OsmAnd favorites with a fixed three-group convention — consistency is what makes markers usable under stress:

- **WATER (blue):** verified sources from the §3.1 hierarchy, tagged with priority number and treatment requirement.
- **MED (red):** urgent care, pharmacies, veterinary suppliers, household members with medical training.
- **RALLY (yellow):** pre-agreed family assembly points (§6.3 defines the selection standard).

▲ BACKUP YOUR MARKERS

Export saved favorites as a standard map-data file (.gpx) and back it up inside 03_Offline_Maps so custom markers can be instantly imported onto the reserve terminal if the primary device fails. Mark points during steady state, not during the event.

SECTION 5

IMMEDIATE MEDICAL RESPONSE

Scope: field stabilization when professional care is delayed or unreachable. Where licensed care is reachable, transport takes priority over every procedure below (§0.5).

5.1 MARCH ASSESSMENT SEQUENCE

SUSTAINED

Assess every casualty in this fixed order — it is ordered by what kills fastest.

LETTER	CHECK	ACTION THRESHOLD
M — Massive bleeding	Visible life-threatening bleeding	Control before anything else — see 5.2
A — Airway	Is the airway open and clear?	Tilt head back, lift chin; recovery position if unconscious and breathing
R — Respiration	Rate and quality of breathing	Normal adult: 12–20 breaths/min. Sustained under 10 or over 30 = escalate
C — Circulation	Pulse, skin color, temperature	Weak/rapid pulse + pale, cool, clammy skin = treat for shock: lie flat, insulate, raise legs unless injured
H — Hypothermia / Head	Body temperature; pupils; alertness	Insulate from ground first, cover second; declining alertness = highest evacuation priority

■ REASSESSMENT CADENCE

Full sequence every **15 minutes** for unstable casualties; every **hour** once stable. Log times and findings on paper — memory fails under stress.

5.2 BLEEDING CONTROL AND WOUND MANAGEMENT

IMMEDIATE

- **Direct pressure** is the first response to all bleeding: firm, continuous, both hands, through a clean dressing. Do not lift to check — **10 minutes minimum by the clock**.
- **Pressure not controlling a limb bleed:** apply a tourniquet 5–7 cm (2–3 in) above the wound — never on a joint — tighten until bleeding stops, and **write the application time** on the band or the casualty's skin. A properly applied tourniquet hurts — that is expected. Once applied, it is removed only when advanced care is reached.
- **Wound care after control:** rinse with treated water (§3.2 — boiled-barrier water for wounds), remove visible debris, dress, and re-check at every MARCH reassessment for spreading redness, heat, swelling, or red streaks moving from the wound.
- **Do not** stitch wounds in the field or pull out deeply embedded objects — pad and secure embedded objects in place and dress around them.

▲ IMPROVISED TOURNIQUET — READ BEFORE YOU NEED IT

Never use makeshift wire, paracord, or thin cords as a tourniquet; they slice through tissue and cause severe nerve damage without reliably stopping deep arterial bleeding. If you must improvise a tourniquet, use a thick cloth band at least 2 inches wide (like a folded triangular bandage or a belt) and a strong twisting stick (like a heavy screwdriver, wrench, or thick wooden dowel). Wrap the band, insert the stick, and twist it until the bleeding completely stops, then tie the stick firmly in place so it cannot unwind.

5.3 ENVIRONMENTAL INJURIES

SUSTAINED

CONDITION	RECOGNITION	FIELD RESPONSE
Heat exhaustion	Heavy sweating, weakness, nausea, headache	Shade, cool water sips, wet cloth to neck/armpits/groin; rest until urine runs pale
Heat stroke	Hot dry or clammy skin, confusion, collapse	Life threat. Cool aggressively — immersion or continuous wet-cloth-and-fanning; highest evacuation priority
Hypothermia	Shivering → slurred speech → shivering stops (critical sign)	Dry clothing, insulate from ground, external warmth, warm sweet liquids only if fully awake; handle gently
Frostbite	Waxy, numb, pale fingers/toes/ears	Rewarm in 37–39 °C (99–102 °F) water only when refreezing is impossible; never rub tissue; never rewarm with direct flame
Immersion foot	Numb, swollen, blotchy feet after sustained wet exposure	Dry, elevate, rewarm gradually; rotating dry socks is the prevention (§7.3)

5.4 WIKIMED LOOKUP PROCEDURE

72 - HR

The WikiMed archive in [01_Kiwix_Offline_Wikis](#) is the deep reference behind this section — full medication dosing tables, child-dose adjustments, and conditions beyond this manual's scope.

- 01 Launch the Kiwix Reader from the directory, then open the WikiMed archive file (verified during §2.4 activation).
- 02 Search by **symptom first, suspected condition second** — symptom pages link onward to possible causes and are faster under stress.
- 03 Read the **Management / Treatment** section before the Diagnosis detail — action first, theory when time permits.
- 04 **Battery discipline:** identify the needed article, read once, copy dosages and steps to paper, screen off. Do not browse.
- 05 Bookmark your household's known conditions, allergies, and medication pages during steady state (§7.2), not during the event.

OFF-GRID COMMUNICATIONS

6.1 COMMUNICATIONS HIERARCHY AND MONITORING WINDOWS

SUSTAINED

Receiving information outranks transmitting it.

PRIORITY	ASSET	ROLE
1	Battery / hand-crank AM-FM radio	Official broadcasts and alerts; the baseline asset
2	NOAA weather radio (162.400–162.550 MHz)	Continuous hazard alerts where coverage exists
3	License-free walkie-talkies (FRS/GMRS)	Household and neighbor coordination, ~1–3 km real-world range
4	Radio scanner (receive-only)	Listening to local emergency services for situational awareness
5	Amateur (ham) two-way radio	Regional reach — transmitting requires a license; listening is legal for everyone

■ MONITORING WINDOWS

Power radios at fixed times — **top of every odd hour for 10 minutes** during the first 72 hours, then **0700 / 1200 / 1900** sustained. All household members know the schedule; radios stay off outside windows. Log every substantive broadcast on paper: time, station, content.

6.2 PROTECTIVE POSTURE (EMP)

72-HR

▲ EMP SHIELD RULE

The Faraday pouch protects **what is inside it at the moment of an electromagnetic surge** — nothing else. Re-seal per §1.3 every time: a gapped fold is an open pouch.

- **Sealed at all times (steady state):** the reserve data drive (§1.4), one walkie-talkie, spare USB adapter set.
- **Sealed during elevated threat:** reserve terminal (§2.2), hand-crank radio at night outside monitoring windows.
- **Never achievable:** the deployed primary terminal is accepted risk while in use. During elevated threat, tighten monitoring windows to 5 minutes.

6.3 FAMILY COMMUNICATION PLAN AND RALLY PROTOCOL

SUSTAINED

Agree on all of the following during steady state; print copies into every household member's kit and

02_Custom_Survival_Checklists:

- 01 **Rally points (§4.4 markers):** RALLY-1 near home (walkable by the youngest member), RALLY-2 outside the neighborhood, RALLY-3 outside the region. Each carries a **hold time** — how long to wait before moving to the next (default: 24 hours).
- 02 **Out-of-region contact:** one relative or friend far enough away to be unaffected. If any communications path exists, every member reports to that single contact — one central check-in point works far better than everyone trying to reach everyone else, and it survives a patchy, partially restored network.
- 03 **Radio protocol:** one agreed walkie-talkie channel and two backups. First-name basis only; no addresses, no inventory, no plans over the air (§3.5 security discipline applies to transmissions too).
- 04 **Message discipline:** transmissions under 30 seconds, on the same monitoring windows as 6.1. Listen for two full minutes before transmitting.
- 05 **No-comms default:** if separated with no contact by RALLY-1's hold time, all members execute the rally sequence in order. **The plan is the communication.**

■ TRANSMISSION DOCTRINE

Listen two minutes. Transmit thirty seconds. Say nothing a stranger should hear.

SECTION 7

SUSTAINMENT CHECKLISTS

Printable duplicates of every list below live in 02_Custom_Survival_Checklists. Work from paper; preserve battery (§3.5).

7.1 THE 72-HOUR ACTION CHECKLIST

HOURL 0 - 1

- ☐ Two-indicator trigger confirmed and activation time logged (§2.1)
- ☐ Faraday seal broken; drive deployed (§1.3)
- ☐ Device power audit complete; primary and reserve terminals assigned (§2.2)
- ☐ Drive mounted; four dashboard tiles verified (§2.4, Fig. 2-A)
- ☐ Refrigerator/freezer marked **DO NOT OPEN** except scheduled meals (§3.5)

HOURL 1 - 6

- ☐ Water inventory: sealed stock + water heater volume totaled against 4 L/person/day (§3.1)
- ☐ Water heater drained to containers (§3.1)
- ☐ Food inventory logged on the paper calorie tracker (§3.5)
- ☐ First radio monitoring window observed and logged (§6.1)
- ☐ Household briefed: rally points, radio channels, monitoring schedule (§6.3)

HOURL 6 - 24

- ☐ OsmAnd verified on primary terminal; position confirmed (§4.1, Fig. 4-A)
- ☐ PACE routes reviewed; paper turn lists staged in kits (§4.3)
- ☐ Water treatment line established for untrusted sources (§3.2)
- ☐ Medical kit staged; MARCH card printed or copied out (§5.1)
- ☐ Protective posture set: reserve assets sealed (§6.2)

HOURL 24 - 72

- ☐ Refrigerated perishables consumed or discarded per the 4-hour rule (§3.5)
- ☐ Daily consumption logged: water, calories, battery percentages (§3.4, §3.5)
- ☐ Monitoring windows held to schedule; broadcast log current (§6.1)
- ☐ Out-of-region contact attempted once per day if any path exists (§6.3)
- ☐ Hour-72 decision: shelter in place vs. movement, using PACE routes (§4.3)

7.2 QUARTERLY MAINTENANCE / DRIVE RE-FLASH SCHEDULE

Steady-state discipline is what makes activation possible. Calendar these — memory is not a system.

CADENCE	TASK	REFERENCE
Quarterly	Mount both drives; verify all four directories open and match the §1.1 manifest	§1.1
Quarterly	Open one WikiMed article via the Kiwix Reader on a current household device	§2.4
Quarterly	Confirm OsmAnd installs and launches on current phones; verify the favorites backup file (.gpx) is current	§4.1, §4.4
Quarterly	Re-verify WATER / MED / RALLY markers still valid; edition check on this manual	§4.4, §0.4
Semi-annual	Rotate stored water; inspect containers	§3.4
Annual	Re-flash both drives with updated archives, map regions, and checklist revisions; verify, then re-seal	§1.4
Annual	Replace the pouch if the fold shows wear; test the seal	§1.3
After any use	Full re-verify + re-seal of both drives; restock consumed checklist printouts	§1.3

⚠ ANNUAL RE-FLASH IS NOT OPTIONAL

Maps age, medical references update, and checklists evolve. A drive that has not been re-flashed in two years is an artifact, not an asset.

7.3 SKILLS PRACTICE CADENCE

■ TRAINING DOCTRINE

Equipment without rehearsed skills is cargo.

CADENCE	DRILL	REFERENCE
Monthly · 15 min	One member runs the §2 activation drill cold — seal to four verified tiles, timed. Target: under 10 minutes	§2
Monthly · 15 min	Water treatment rehearsal — one liter through the three-barrier standard, bleach dosing from memory, verified against the printed card	§3.2
Quarterly	Family radio check on the agreed channels including both backups; message discipline enforced	§6.3
Quarterly	Navigation walk — one PACE route segment walked with the paper turn list, device used for verification checks only	§4.2
Semi-annual	Full rally drill — household reaches RALLY-1 unannounced; debrief hold-time decisions	§6.3
Annual	MARCH talk-through with a simulated casualty; tourniquet placement practiced on a practice dummy or padded object — never on a person's bare limb at full tension	§5.1, §5.2

DRILL LOG — EVERY DRILL, EVERY TIME

- ☐ Date and participants recorded
- ☐ Time-to-complete recorded against target
- ☐ One improvement item identified and written down
- ☐ Log filed in 02_Custom_Survival_Checklists

7.4 FIELD OPERATIONAL TRAINING DRILLS

Run these drills on the §7.3 cadence. Each is executed as a household, timed, and logged. A drill that ends in an argument instead of a debrief still counts — that is what the debrief is for.

BLOCK A: SYSTEM DIAGNOSTIC DRILLS

- 01 **Drill 01: The 30-Minute Comms Blackout** — Kill structural internet/cell access. Deploy the auxiliary battery/hand-crank radio, sweep AM/NOAA bands to locate the strongest emergency broadcast, and log the frequency.
- 02 **Drill 02: The Gravity-Assisted Flush Trial** — Shut down the residential water input valve. Pour one gallon of secondary non-potable water rapidly into the toilet bowl to trigger a manual siphon flush and verify baseline sanitization volume.
- 03 **Drill 03: The Offline Vector Pathfinding Exercise** — Place the terminal in total airplane mode. Launch the local map database, plot a non-vehicular route to a rally location 2 miles away, and identify three high-ground landmarks entirely from cached elevation layers.

BLOCK B: HOUSEHOLD OPERATION DRILLS

- 04 **Drill 04: The 60-Minute 'Breaker-Off' Dinner** — Flip the main residential electrical breaker for 1 hour during dinner prep. Navigate the structure, deploy blackout lighting, and execute a meal plan completely grid-free.
- 05 **Drill 05: The 'Analog Direct' Meetup** — Set a rally location 2 miles away. Family members depart separately with zero active cellular voice, text, or data usage. Navigate and cross-rally using only offline maps and pre-set visual parameters.
- 06 **Drill 06: The Core Kit Siphon Trial** — Task every household member with manually resetting a static toilet bowl back to operational status using exactly one gallon of stored utility water.

■ DRILL STANDARD

Every drill is logged per §7.3: date, participants, time-to-complete, one improvement item. An undrilled protocol is a theory.

DIY VAULT BUILD COMPENDIUM

Complete blueprint for replicating the Grid Down Data Drive from scratch — hardware, open-source data, and directory architecture. Build during steady state; verify per §7.2 before you depend on it.

8.1 HARDWARE SOURCING SPECIFICATIONS

Two components. Do not economize on either — the entire system rides on them.

COMPONENT	REQUIRED SPECIFICATION
USB flash drive	Ruggedized, IP68 waterproof , USB-C (with USB-A capability via adapter or dual connector). Capacity 128 GB minimum — maps and wiki archives grow with every annual re-flash (§7.2). Metal unibody shell; operating range -20°C to 60°C .
Faraday pouch	Dual-fold closure, military-grade shielding fabric (look for tested attenuation ratings and a double-layer construction). Sized to hold the drive, USB adapters, and a spare cable with the fold fully closed. A pouch that barely closes will be a pouch that gets left open.

▲ SOURCING RULES

Buy from established vendors with verifiable specification sheets — marketplace listings that name no shielding rating or waterproof standard are decorative, not protective. **Test before you trust:** submerge the drive (sealed cap) in a bowl of water for 30 minutes, dry, then verify it mounts and all files read. Place a phone inside the sealed pouch and call it — a ringing phone is a failed pouch.

■ BUILD DOCTRINE

Two drives, two pouches, two locations (§1.4). Build both in the same session so both copies are identical.

8.2 DATA SOURCING DIRECTORY

Every library on the drive is open-source and free to download. Complete all downloads during steady state, on a reliable connection — the archives are large.

01 KIWIX OFFLINE WIKIS — for 01_KIWIX_OFFLINE_WIKIS

- 01 Download the **Kiwix Reader** software for every platform your household uses (Windows, macOS, Android) from the official Kiwix project site. Store each installer on the drive itself — the reader must travel with the data.
- 02 From the official **Kiwix library** (library.kiwix.org), download the **WikiMed medical archive** in compressed **.zim** format, plus a general-reference ZIM if capacity allows.
- 03 Verify each ZIM after download: open it in the Kiwix Reader and confirm articles render before copying to the drive.

03 OFFLINE MAPS — for 03_OFFLINE_MAPS

- 01 From the open-source **OsmAnd database** (osmand.net downloads), pull the vector region files (**.obf**) for your country/state — your home region plus every adjacent region (§4.1).
- 02 Download matching **elevation / contour data sets** for the same regions — terrain matters when roads do not.
- 03 Add the OsmAnd Android installer (**.apk**) to **03_Offline_Maps/apk/** so Android devices can install with zero network (§4.1). iOS users: install the app from the App Store now, while the internet exists.

▲ VERIFICATION DRILL — BEFORE FIRST SEAL

Run the full §2 activation procedure on your freshly built drive: mount, dashboard, one WikiMed article, one map region with GPS lock. **A vault you have never opened is a vault you cannot trust.**

8.3 ARCHITECTURE & DIRECTORY STRUCTURE

Create the four directories on a blank, freshly formatted drive **exactly as shown** — names, numbers, and underscores included. The offline HTML dashboard resolves its tile links relative to these paths; a renamed directory is a dead tile.

```
GRIDDOWN DRIVE (root)
|
├── 00_README_and_Blueprint/
|   ├── START_HERE.html      ← dashboard entry point (§2.3)
|   ├── GridDown-Data-Blueprint.pdf
|   └── quick-reference/      ← printable cards
|
├── 01_Kiwix_Offline_Wikis/
|   ├── kiwix-reader/        ← installers, every platform
|   ├── wikimed.zim
|   └── reference.zim
|
├── 02_Custom_Survival_Checklists/
|   ├── water/  medical/  food/
|   └── maintenance-logs/
|
└── 03_Offline_Maps/
    ├── apk/                 ← OsmAnd installer (Android)
    ├── regions/             ← .obf vector files (§4.1)
    ├── elevation/           ← contour / terrain sets
    └── favorites-backup.gpx ← marker backup (§4.4)
```

FINAL BUILD VERIFICATION — RUN ON BOTH COPIES

- ☐ Four directories present, spelled exactly as above, in order 00–03
- ☐ `START_HERE.html` opens from the drive and all four dashboard tiles resolve (§2.4)
- ☐ One WikiMed article renders in the Kiwix Reader from the drive copy
- ☐ Home region map opens in OsmAnd; GPS position confirmed outdoors
- ☐ Submersion and pouch tests passed (8.1); both copies sealed per §1.3
- ☐ Build date logged; first quarterly verification calendared (§7.2)

■ COMPLETION STANDARD

The build is complete when the §2 activation drill runs clean on both copies — not before.

QUICK-REFERENCE CARD

GRID DOWN ACTIVATION – 10 LINES

- 01 Two indicators, two hours → **break the seal** (\$2.1)
- 02 Best-charged phone = **primary terminal**; airplane ON, GPS ON (\$2.2)
- 03 Second device **OFF** and sealed as reserve (\$2.2)
- 04 Mount drive → 00_README_and_Blueprint → START_HERE.html (\$2.3)
- 05 **Four tiles verified** = system nominal (\$2.4)
- 06 Water first: **4 L/person/day**; filter → bleach 2 drops/L → boil 1 min (\$3)
- 07 Fridge shut; **1,200–1,500 kcal/day**; 4-hour rule on perishables (\$3.5)
- 08 OsmAnd: install → copy regions → **confirm GPS position outdoors** (\$4)
- 09 Bleeding: **pressure 10 min by the clock**; MARCH order always (\$5)
- 10 Radio windows: **odd hours ×10 min**; listen 2 min before transmitting (\$6)

WATER OUTRANKS FOOD. RECEIVING OUTRANKS TRANSMITTING. PAPER OUTRANKS PIXELS.